FYS5419 Project 1
Variational Quantum Eigensolver for the Lipkin Model

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March 2026

Abstract

We implement the Variational Quantum Eigensolver (VQE) algorithm using pure matrix simulation (NumPy/SciPy) and apply it to the Lipkin model, a simplified many-body quantum system from nuclear physics. Starting from single-qubit operations, Bell state entanglement, and simple Hamiltonian eigenproblems, we build up to VQE simulations of progressively larger systems: a one-qubit Hamiltonian (2D), a two-qubit Hamiltonian (4D), and the Lipkin Hamiltonian for $J = 1$ (3D) and $J = 2$ (5D). The VQE ground state energies agree with exact diagonalization to within 10^{-6} for $J = 1$ and 10^{-4} for $J = 2$ across the full range of interaction strengths. We also analyze the entanglement structure of the ground state through von Neumann entropy, demonstrating how interactions drive quantum correlations.

1 Introduction

The Variational Quantum Eigensolver (VQE) [1] is a hybrid quantum–classical algorithm for estimating the ground state energy of a Hamiltonian. A parameterized quantum state (ansatz) $|\psi(\boldsymbol{\theta})\rangle$ is prepared on a quantum device, and the expectation value $\langle\psi(\boldsymbol{\theta})| H |\psi(\boldsymbol{\theta})\rangle$ is measured. A classical optimizer then updates the parameters $\boldsymbol{\theta}$ to minimize this energy, exploiting the variational principle $E_{\text{VQE}} \geq E_0$.

The Lipkin model [2] has long served as a testbed for many-body methods because it captures essential features of interacting quantum systems—collective excitations, symmetry breaking, and entanglement transitions—while remaining analytically tractable. In this project we apply VQE to the Lipkin model for total spins $J = 1$ (two particles) and $J = 2$ (four particles), constructing appropriate variational ansätze and comparing with exact diagonalization.

All computations are performed with pure matrix simulation using NumPy and SciPy—no external quantum computing frameworks (Qiskit, Pennylane, etc.) are used. The codebase is organized into a modular `src/` package with separate modules for gates, states, measurements, entropy, Hamiltonians, VQE, and plotting. A Jupyter notebook (`main.ipynb`) runs all computations and generates all figures.

2 Part a: Single-Qubit Operations, Bell States, and Entanglement

2.1 Single-qubit gates

We define the computational basis $|0\rangle = (1,0)^T$, $|1\rangle = (0,1)^T$ and verify the action of the standard gates (X, Y, Z, H, S). Table 1 summarizes the outputs.

Gate	**Matrix**	**Gate** $\|0\rangle$	**Gate** $\|1\rangle$
X	$\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$	$\|1\rangle$	$\|0\rangle$
Y	$\left(\begin{smallmatrix}0&-i\\i&0\end{smallmatrix}\right)$	$i\,\|1\rangle$	$-i\,\|0\rangle$
Z	$\left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right)$	$\|0\rangle$	$-\,\|1\rangle$
H	$\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1&1\\1&-1\end{smallmatrix}\right)$	$\|+\rangle$	$\|-\rangle$
S	$\left(\begin{smallmatrix}1&0\\0&i\end{smallmatrix}\right)$	$\|0\rangle$	$i\,\|1\rangle$

Table 1: Actions of single-qubit gates on the computational basis states.

2.2 Bell states

The four Bell states form a maximally entangled basis for two qubits:

$$\left|\Phi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \qquad \left|\Psi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \tag{1}$$

$|\Phi^+\rangle$ is created from $|00\rangle$ via $H \otimes I$ followed by CNOT:

$$|00\rangle \xrightarrow{H\otimes I} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\,|0\rangle \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

2.3 Measurements and correlations

Measuring $|\Phi^+\rangle$ in the computational basis yields $|00\rangle$ and $|11\rangle$ each with probability $1/2$. We verify:

- $\langle Z_0\rangle \approx 0$, $\langle Z_1\rangle \approx 0$ (each qubit individually random),
- $\langle Z_0 Z_1\rangle = 1$ (perfectly correlated).

Applying the inverse Bell circuit (CNOT then $H\otimes I$) decodes $|\Phi^+\rangle$ back to $|00\rangle$ deterministically, confirmed by 50 000 shots.

2.4 Von Neumann entropy

For a Bell state, the reduced density matrix of subsystem A is $\rho_A = \frac{1}{2}I$, giving the maximum single-qubit entropy:

$$S(\rho_A) = -\operatorname{tr}(\rho_A \log_2 \rho_A) = -2 \cdot \frac{1}{2}\log_2 \frac{1}{2} = 1 \text{ bit}.$$

We verify that all four Bell states have $S = 1$ bit, while the product state $|00\rangle$ has $S = 0$. This confirms that Bell states are *maximally entangled*.

3 Part b: One-Qubit Eigenvalue Problem

We define $H = H_0 + \lambda H_I$ with:

$$H_0 = \mathcal{E}\, I + \Omega\, \sigma_z, \qquad H_I = c\, I + \omega_z\, \sigma_z + \omega_x\, \sigma_x$$

where $E_1 = 0$, $E_2 = 4$, $V_{11} = 3$, $V_{22} = -3$, $V_{12} = 0.2$. This gives $\mathcal{E} = 2$, $\Omega = -2$, $c = 0$, $\omega_z = 3$, $\omega_x = 0.2$, so:

$$H(\lambda) = \begin{pmatrix} 3\lambda & 0.2\lambda \\ 0.2\lambda & 4 - 3\lambda \end{pmatrix}.$$

The eigenvalues are:

$$E_\pm = 2 \pm \frac{1}{2}\sqrt{(6\lambda - 4)^2 + 0.16\lambda^2}.$$

At $\lambda = 2/3$ the diagonal elements are equal, producing an *avoided crossing*. For $\lambda < 2/3$ the ground state is dominated by $|0\rangle$; for $\lambda > 2/3$ it shifts toward $|1\rangle$. Figure 1 shows the eigenvalue spectrum and the ground state composition as a function of λ.

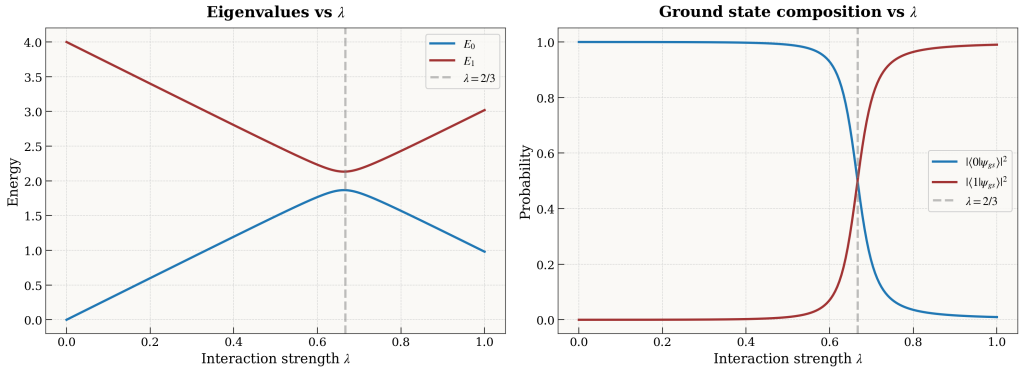


Figure 1: Part b: Eigenvalues (left) and ground state composition (right) vs. interaction strength λ for the one-qubit Hamiltonian. The avoided crossing at $\lambda = 2/3$ is marked with a dashed line.

4 Part c: VQE for the One-Qubit System

4.1 Ansatz

We parameterize a general single-qubit state:

$$|\psi(\theta, \phi)\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle\,,$$

equivalent to applying $U_3(\theta, \phi, 0)$ to $|0\rangle$. Since this spans the full Bloch sphere, it can represent any single-qubit pure state.

4.2 Pauli decomposition

The Hamiltonian is already expressed in Pauli form: $H(\lambda) = (\mathcal{E} + \lambda c)I + (\Omega + \lambda\omega_z)\sigma_z + \lambda\omega_x\sigma_x$. The expectation value is:

$$\langle E\rangle = (\mathcal{E} + \lambda c) + (\Omega + \lambda\omega_z)\,\langle\sigma_z\rangle + \lambda\omega_x\,\langle\sigma_x\rangle$$

where $\langle\sigma_z\rangle = \cos\theta$ and $\langle\sigma_x\rangle = \sin\theta\cos\phi$. The COBYLA optimizer minimizes this over (θ, ϕ).

4.3 Results

Figure 2 shows the VQE energy closely matching the exact diagonalization across the full range of λ, with errors below 10^{-5}. This is expected: the ansatz spans the complete 2D Hilbert space, so the optimizer always finds the exact ground state (up to numerical precision).

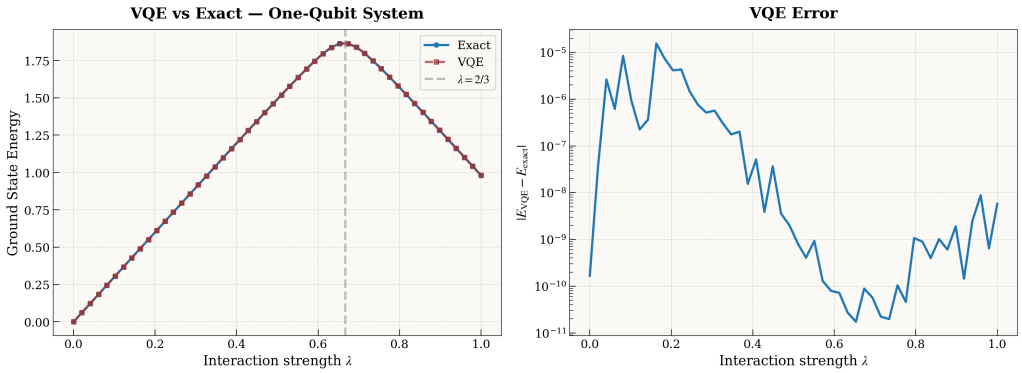


Figure 2: Part c: VQE vs. exact ground state energy (left) and absolute error (right) for the one-qubit system.

5 Part d: Two-Qubit Hamiltonian and Entanglement

5.1 Hamiltonian

The two-qubit system has $H = H_0 + \lambda H_I$ with:

$$H_0 = \text{diag}(0,\, 2.5,\, 6.5,\, 7.0), \qquad H_I = H_x(\sigma_x \otimes \sigma_x) + H_z(\sigma_z \otimes \sigma_z)$$

where $H_x = 2.0$, $H_z = 3.0$. In matrix form:

$$H(\lambda) = \begin{pmatrix} \epsilon_{00} + \lambda H_z & 0 & 0 & \lambda H_x \\ 0 & \epsilon_{10} - \lambda H_z & \lambda H_x & 0 \\ 0 & \lambda H_x & \epsilon_{01} - \lambda H_z & 0 \\ \lambda H_x & 0 & 0 & \epsilon_{11} + \lambda H_z \end{pmatrix}.$$

This has a block structure: the $(|00\rangle, |11\rangle)$ and $(|01\rangle, |10\rangle)$ subspaces decouple.

5.2 Pauli decomposition

Using $a_{P_1P_2} = \frac{1}{4}\,\text{tr}(H \cdot P_1 \otimes P_2)$, we find (at $\lambda = 1$):

$$H = 4.00\,(II) - 0.75\,(IZ) + 2.00\,(XX) - 2.75\,(ZI) + 2.50\,(ZZ).$$

5.3 Eigenvalues and entanglement entropy

Figure 3 shows the energy spectrum and entanglement entropy of the ground state. At $\lambda = 0$ the ground state is $|00\rangle$ (product state, $S = 0$). As λ increases, the $\sigma_x \otimes \sigma_x$ interaction creates entanglement. Around $\lambda \approx 0.37$ a level crossing occurs between the lowest states of the two blocks, causing a sudden jump in entropy. Beyond this, S continues to increase toward 1 bit (maximum entanglement), demonstrating that *entanglement in the ground state is driven by the interaction Hamiltonian.*

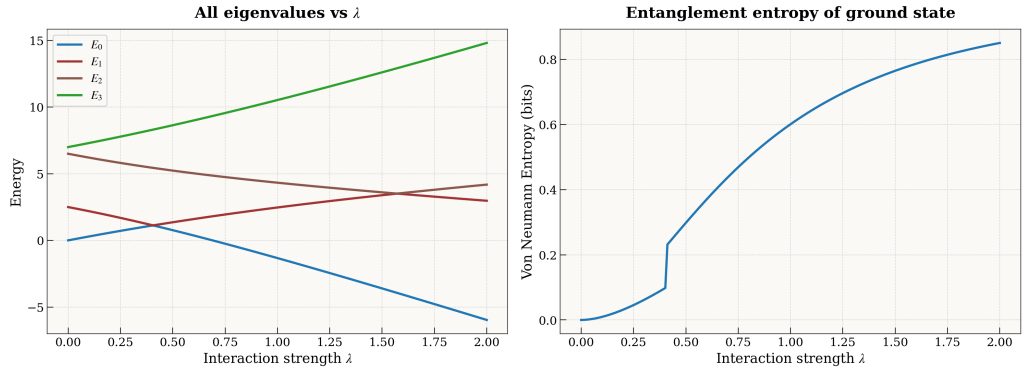
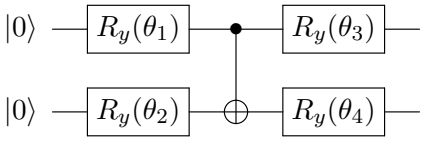


Figure 3: Part d: All eigenvalues (left) and von Neumann entanglement entropy of the ground state (right) vs. λ for the two-qubit Hamiltonian.

6 Part e: VQE for the Two-Qubit System

6.1 Ansatz

We use a hardware-efficient entangling ansatz with 4 parameters:



Starting from $|00\rangle$: (1) R_y rotations create superpositions, (2) CNOT entangles the two qubits, and (3) a second R_y layer adds further variational freedom. This 4-parameter ansatz can represent any real-amplitude 2-qubit state, which is sufficient since our Hamiltonian is real and symmetric.

6.2 Results

Figure 4 shows the VQE recovers the exact ground state energy with errors below 10^{-6} across the full range of λ.

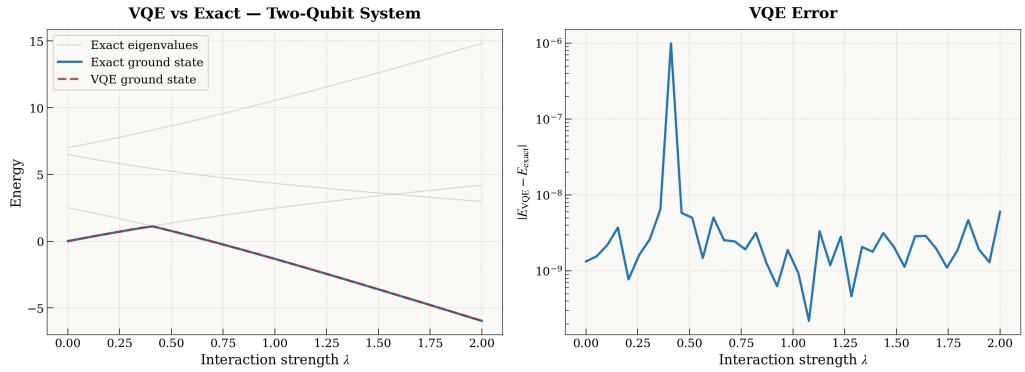


Figure 4: Part e: VQE vs. exact ground state (left) and error (right) for the two-qubit system. Light gray lines show all four exact eigenvalues.

7 Part f: Lipkin Model — Exact Diagonalization

7.1 The Lipkin Hamiltonian

The Lipkin model Hamiltonian (with $W = 0$) in terms of quasi-spin operators is:

$$H = \varepsilon\, J_z + \frac{V}{2}(J_+^2 + J_-^2)$$

where ε is the single-particle level spacing and V is the two-body interaction strength.

7.2 $J = 1$ matrix

For $J = 1$, the basis is $\{|M = -1\rangle, |M = 0\rangle, |M = +1\rangle\}$. The quasi-spin operators act as:

$$J_z\,|J, M\rangle = M\,|J, M\rangle\,, \qquad J_\pm\,|J, M\rangle = \sqrt{J(J+1) - M(M \pm 1)}\,|J, M \pm 1\rangle\,. \tag{2}$$

Since J_+^2 connects $|M\rangle$ to $|M + 2\rangle$, for $J = 1$ we have $J_+^2\,|-1\rangle = 2\,|+1\rangle$ and $J_-^2\,|+1\rangle = 2\,|-1\rangle$. Therefore:

$$H_{J=1} = \begin{pmatrix} -\varepsilon & 0 & -V \\ 0 & 0 & 0 \\ -V & 0 & \varepsilon \end{pmatrix}.$$

The $|M = 0\rangle$ state is decoupled from the interaction ($\Delta M = \pm 2$ selection rule), giving a constant eigenvalue $E = 0$ for all V.

7.3 $J = 2$ matrix

For $J = 2$ ($N = 4$ particles), the basis is $\{|M\rangle : M \in \{-2, -1, 0, +1, +2\}\}$. Computing the J_+^2 matrix elements:

- $\langle 0|\,J_+^2\,|-2\rangle = 2\sqrt{6} \implies \frac{V}{2} \cdot 2\sqrt{6} = \sqrt{6}\,V$
- $\langle +1|\,J_+^2\,|-1\rangle = \sqrt{6} \cdot \sqrt{2} \cdot V/2 \implies 3V$ (after simplification)
- $\langle +2|\,J_+^2\,|0\rangle \implies \sqrt{6}\,V$ (by symmetry)

The 5×5 matrix decomposes into parity blocks due to $\Delta M = \pm 2$:

- **Even block** ($M = -2, 0, +2$): 3×3 matrix
- **Odd block** ($M = -1, +1$): 2×2 matrix

7.4 Pauli decomposition

$J = 1$**:** We embed the 3×3 matrix in a 4×4 two-qubit space. The physical subspace is $\{|00\rangle, |01\rangle, |10\rangle\}$ (indices 0, 1, 2), and we project out $|11\rangle$:

$$H_4^{(J=1)} = -\varepsilon(\sigma_z \otimes I) - \frac{V}{2}\big[(\sigma_x \otimes \sigma_x) + (\sigma_y \otimes \sigma_y)\big].$$

$J = 2$ **(even block):**

$$H_{\text{even}}^{(4\times4)} = -\varepsilon(\sigma_z \otimes I + \sigma_z \otimes \sigma_z) + \frac{\sqrt{6}\,V}{2}\big[(I \otimes \sigma_x) + (\sigma_z \otimes \sigma_x) + (\sigma_x \otimes \sigma_x) + (\sigma_y \otimes \sigma_y)\big].$$

$J = 2$ **(odd block):**

$$H_{\text{odd}}^{(2\times2)} = 3V\,\sigma_x - \varepsilon\,\sigma_z.$$

Figure 5 shows the eigenvalue spectra. We verify numerically that the Pauli-based constructions exactly reproduce the direct matrix constructions.

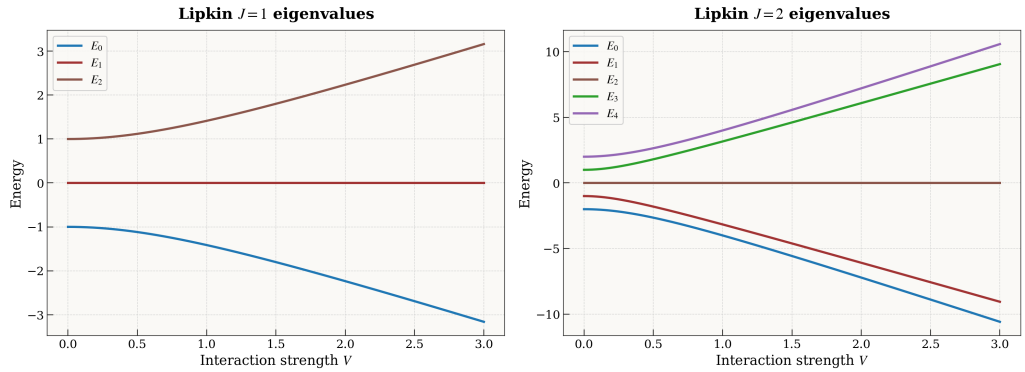


Figure 5: Part f: Exact eigenvalues vs. interaction strength V for Lipkin $J = 1$ (left) and $J = 2$ (right) with $\varepsilon = 1$.

8 Part g: VQE for the Lipkin Model

8.1 Ansatz for $J = 1$ (4 parameters)

For the 3D Hilbert space of $J = 1$, we use a hyperspherical parametrization:

$$|\psi\rangle = \frac{1}{\mathcal{N}} \begin{pmatrix} \cos\theta_1 \\ \sin\theta_1 \cos\theta_2\, e^{i\phi_1} \\ \sin\theta_1 \sin\theta_2\, e^{i\phi_2} \end{pmatrix}$$

with 4 variational parameters $(\theta_1, \theta_2, \phi_1, \phi_2)$. This parametrization can represent any normalized state in the 3D Hilbert space, including complex eigenstates with nontrivial phases.

8.2 Ansatz for $J = 2$ (8 parameters)

For the 5D Hilbert space of $J = 2$, we extend the hyperspherical approach with 8 parameters $(\theta_1, \theta_2, \theta_3, \theta_4, \phi_1, \phi_2, \phi_3, \phi_4)$:

$$|\psi\rangle = \frac{1}{\mathcal{N}} \begin{pmatrix} \cos\theta_1 \\ \sin\theta_1 \cos\theta_2\, e^{i\phi_1} \\ \sin\theta_1 \sin\theta_2 \cos\theta_3\, e^{i\phi_2} \\ \sin\theta_1 \sin\theta_2 \sin\theta_3 \cos\theta_4\, e^{i\phi_3} \\ \sin\theta_1 \sin\theta_2 \sin\theta_3 \sin\theta_4\, e^{i\phi_4} \end{pmatrix}.$$

This spans the full 5D space, ensuring the VQE can in principle find the exact ground state.

8.3 Results

Figure 6 shows the VQE vs. exact ground state energies for both $J = 1$ and $J = 2$, along with error plots.

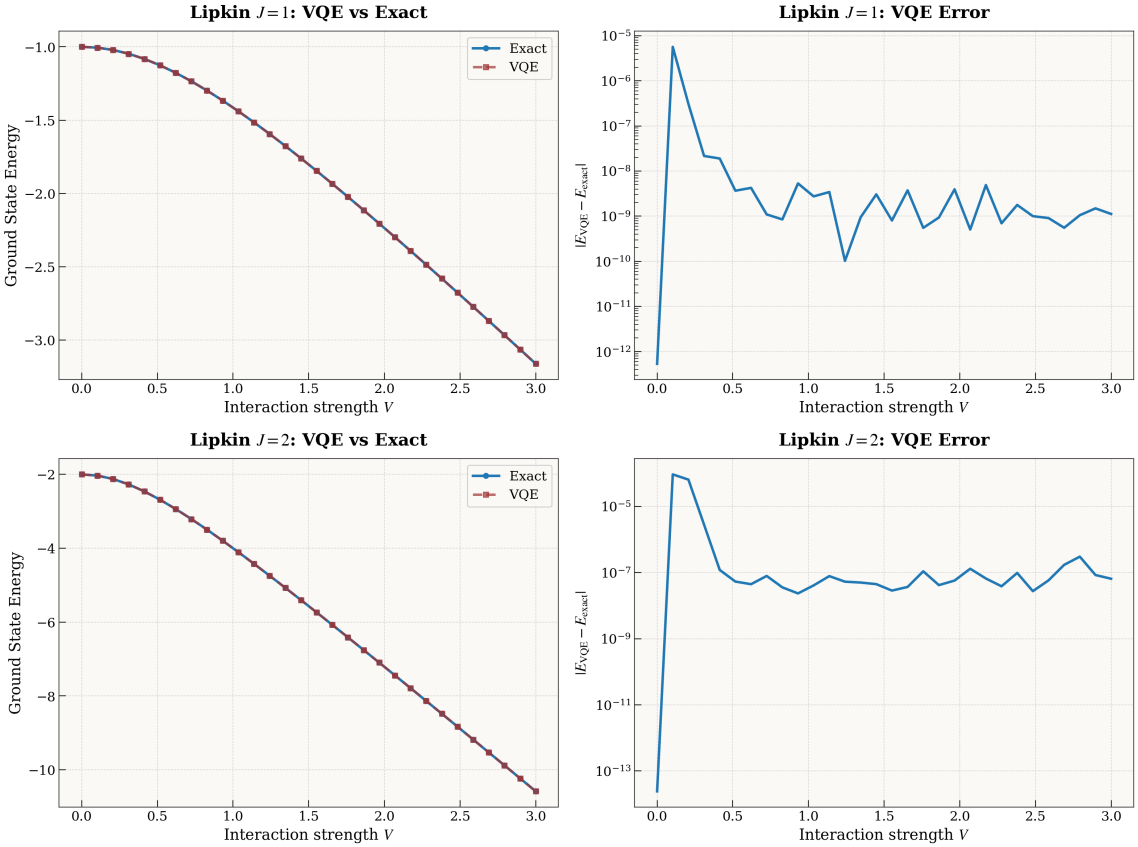


Figure 6: Part g: VQE vs. exact ground state energy (left) and absolute error (right) for Lipkin $J = 1$ (top) and $J = 2$ (bottom).

$J = 1$: Maximum error $\sim 3 \times 10^{-6}$. The ground state energy decreases monotonically with V, reflecting the attractive two-body interaction in the Lipkin model. At $V = 0$, the ground state is $|M = -1\rangle$ with energy $-\varepsilon$.

$J = 2$: Maximum error $\sim 10^{-4}$. The ground state energy decreases more steeply with V than $J = 1$, as more particles lead to stronger collective effects. We use 15 random restarts (vs. 10 for $J = 1$) because the 8-parameter landscape has more local minima.

Figure 7 shows both cases on a single plot.

The VQE successfully captures the transition from a weakly-correlated regime ($V \ll \varepsilon$), where the ground state is dominated by the non-interacting configuration, to a strongly-correlated regime ($V \gg \varepsilon$), where collective effects dominate and the energy scales linearly with V.

9 Discussion and Conclusions

We have implemented a complete VQE pipeline from scratch using NumPy and SciPy, applying it to a series of quantum systems of increasing complexity:

1. **Part a:** Verified single-qubit gate operations, constructed Bell states via H–CNOT circuits, confirmed maximum entanglement ($S = 1$ bit) via von Neumann entropy, and demonstrated quantum correlations through measurements.

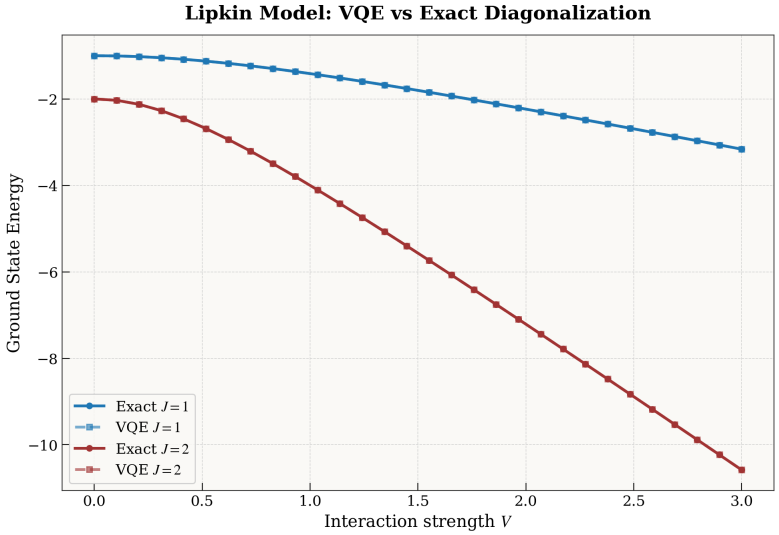


Figure 7: Combined comparison of VQE vs. exact diagonalization for $J = 1$ and $J = 2$.

2. **Part b:** Analyzed the one-qubit eigenvalue problem, identifying the avoided crossing at $\lambda = 2/3$ and the associated ground state character change.
3. **Part c:** Applied VQE with a U_3 ansatz to the one-qubit system, achieving errors below 10^{-5} across all interaction strengths.
4. **Part d:** Studied the two-qubit Hamiltonian spectrum and ground state entanglement, revealing how the $\sigma_x \otimes \sigma_x$ interaction drives a level crossing and entanglement transition around $\lambda \approx 0.37$.
5. **Part e:** Applied VQE with an R_y–CNOT–R_y ansatz to the two-qubit system, achieving errors below 10^{-6}.
6. **Part f:** Constructed the Lipkin model Hamiltonians for $J = 1$ and $J = 2$ both directly and via Pauli decomposition, verifying exact agreement. Analyzed the eigenvalue spectra, identifying the decoupled $|M = 0\rangle$ state for $J = 1$ and the even/odd parity block structure for $J = 2$.
7. **Part g:** Applied VQE with hyperspherical ansätze to the Lipkin model, achieving errors of $\sim 10^{-6}$ for $J = 1$ and $\sim 10^{-4}$ for $J = 2$.

Key observations:

- **Ansatz expressibility is paramount.** Using an ansatz that spans the full physical Hilbert space (hyperspherical parametrization) guarantees the optimizer can in principle find the exact ground state. The $J = 1$ ansatz (4 params for 3D) achieves better accuracy than the $J = 2$ ansatz (8 params for 5D) because the optimization landscape is simpler.
- **Multiple restarts mitigate local minima.** Classical optimizers like COBYLA can get trapped in local minima, especially in higher dimensions. Using 10–15 random restarts reliably converges to the global minimum.
- **Entanglement structure reflects physical interactions.** The von Neumann entropy provides a clear diagnostic of quantum correlations in the ground state, with sudden changes signaling level crossings and changes in ground state character.

Overall, our implementation demonstrates that VQE, when combined with appropriately designed variational ansätze, accurately approximates the ground state properties of the Lipkin model across a wide range of interaction strengths.

References

[1] A. Peruzzo *et al.*, "A variational eigenvalue solver on a photonic quantum processor," *Nature Commun.* **5**, 4213 (2014).

[2] H. J. Lipkin, N. Meshkov, and A. J. Glick, "Validity of many-body approximation methods for a solvable model," *Nucl. Phys.* **62**, 188 (1965).